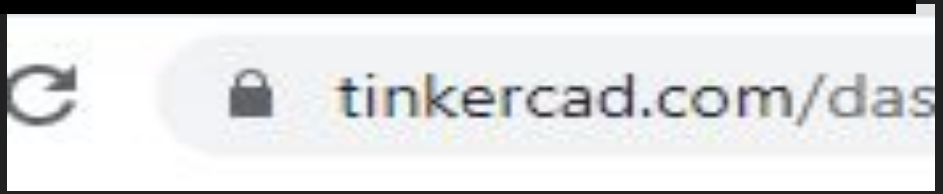
Digital Electronics Interfacing

Lesson 1

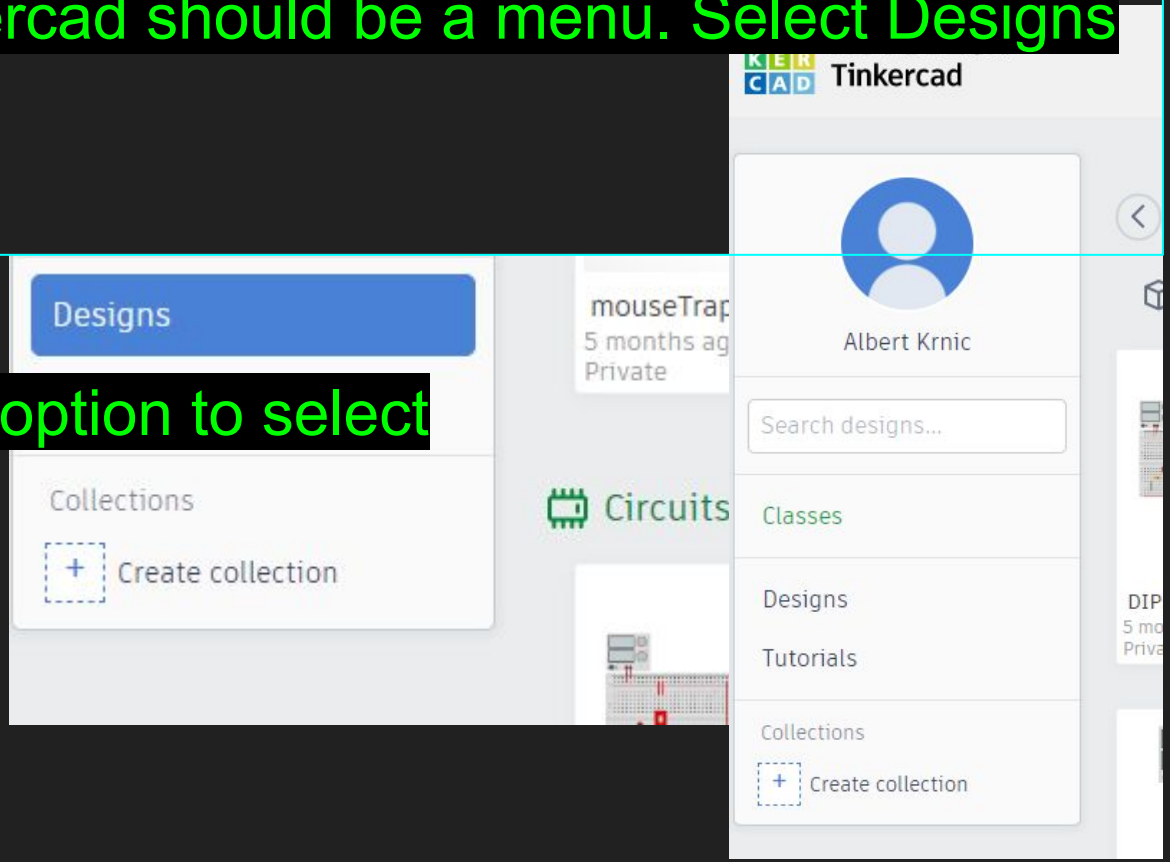
Tinkercad

Create an account at tinkercad.com.



Creating Circuits

On the left side of tinkercad should be a menu. Select Designs from the menu.



You should then see an option to select Circuits.

Click on the link



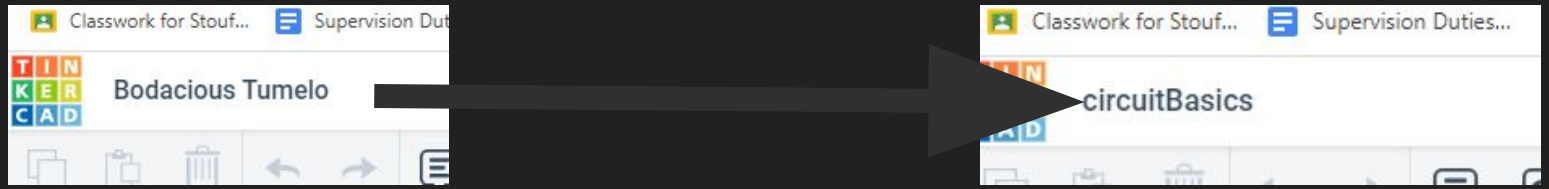
Creating Circuits continued…



To create a new circuit click the +New button from the top right corner of the screen

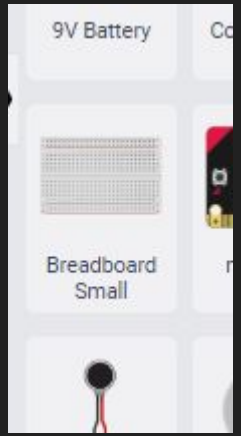
Tinkercad Circuit Naming

You should always give your new circuits a new name. To do this simply click in the current name found in the top left corner of the screen and type in the new name:



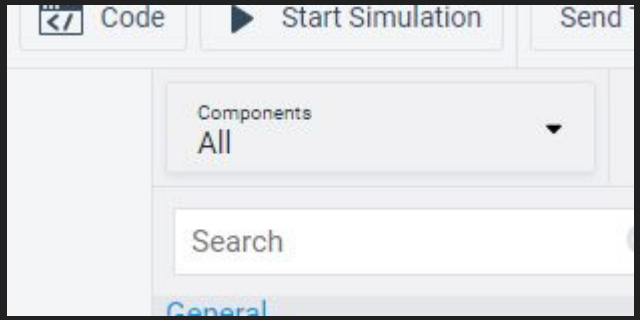
Components

You should have noticed on the right side of the screen is a list of items you can use to start building your circuits. The most important one is the breadboard:

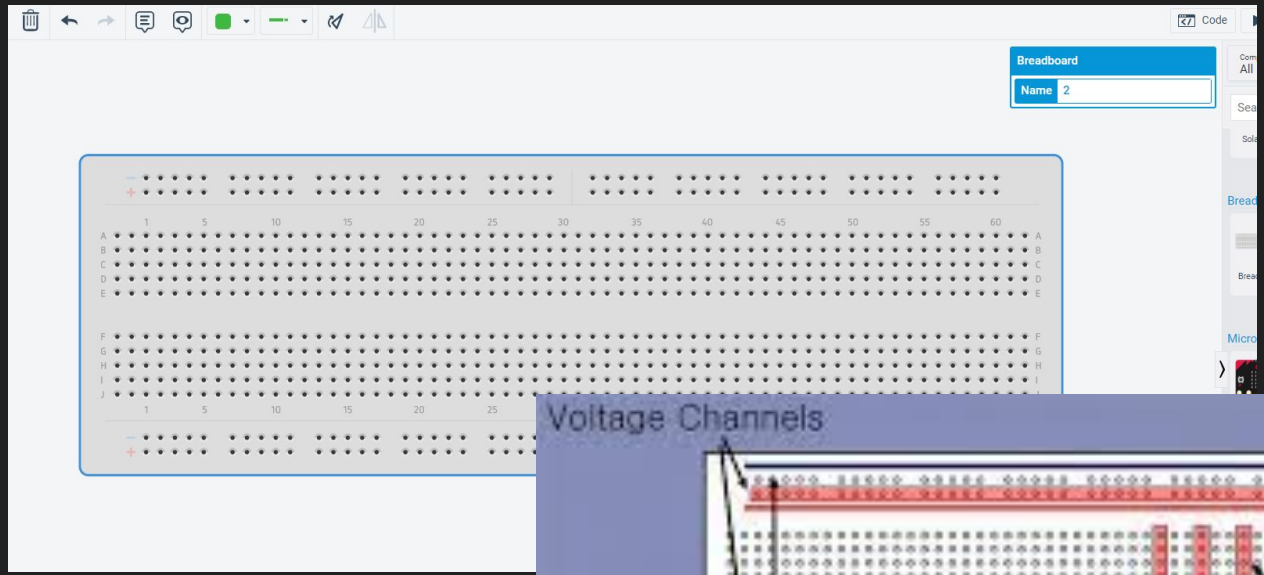


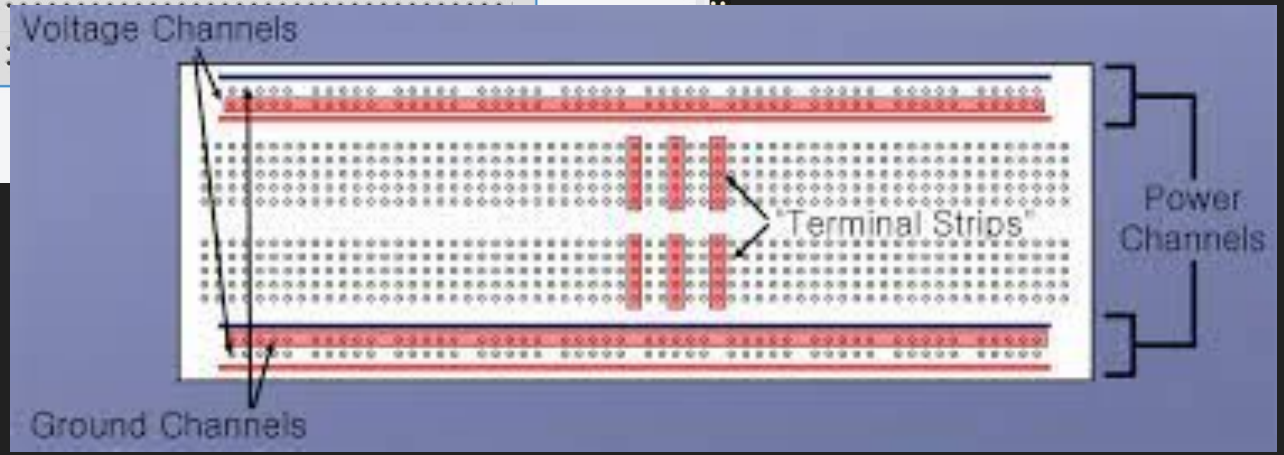
More Components

To get access to many other components look towards the top of the components list and you will find a drop down menu that defaults to Basic. Click and select All.



Add the Large Breadboard-Voltage/Power Channels



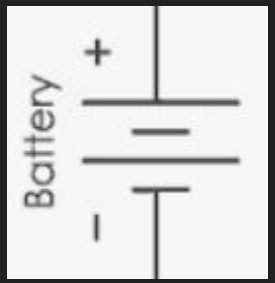


Electricity-Power Source and Ground

An electric current is simply the movement of electrons. These electrons are supplied by a power source i.e. a battery. All electrons want to move to a place called Ground very, very badly!

Schematic Symbols

Note how the sides are labelled + and -. Normally the - side is considered Ground and + is Power.

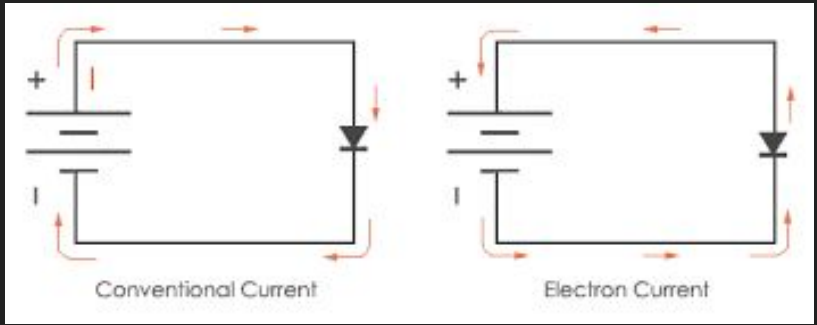




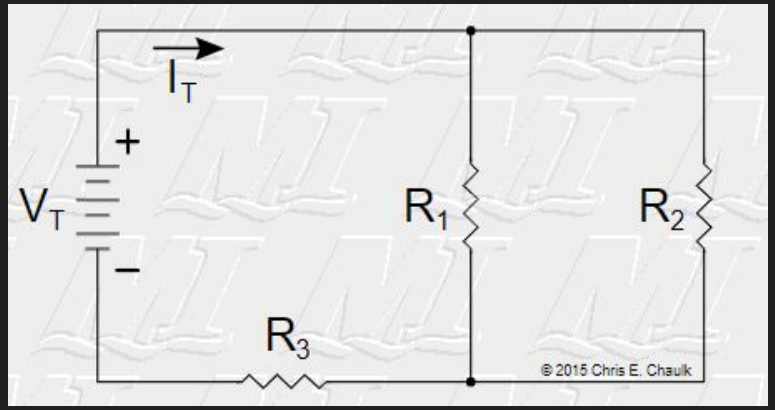
Current Flow-Conventional and Electron Flow

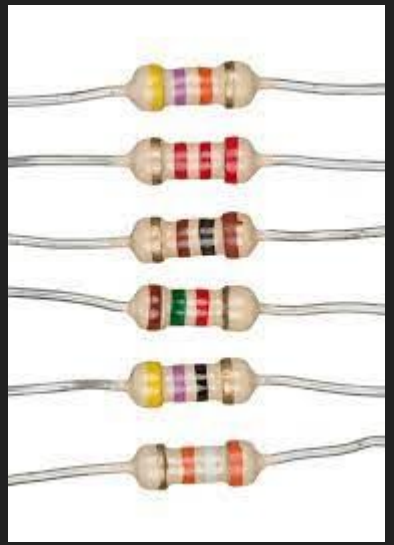
Conventional Current: electrons flow from the + side of a battery/power source to the - side/ground.

Electron Flow: They flow from the - to the +.

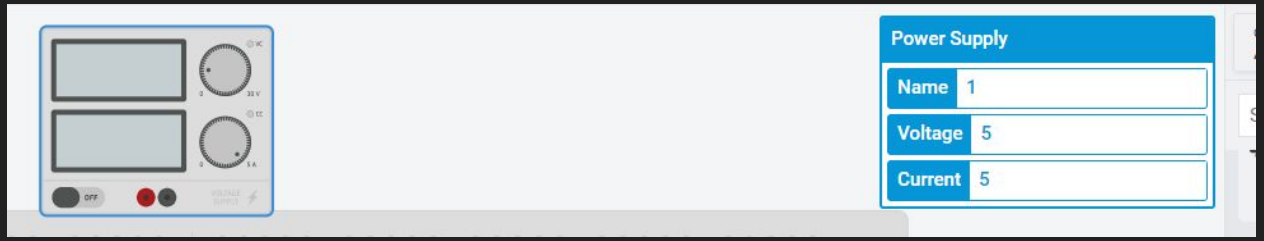


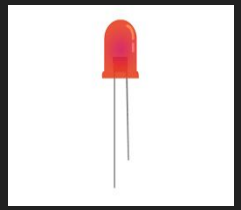
Typical Schematic Circuit-Resistors

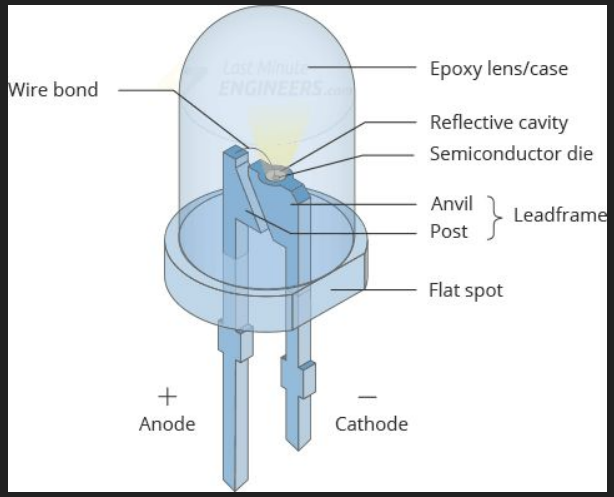


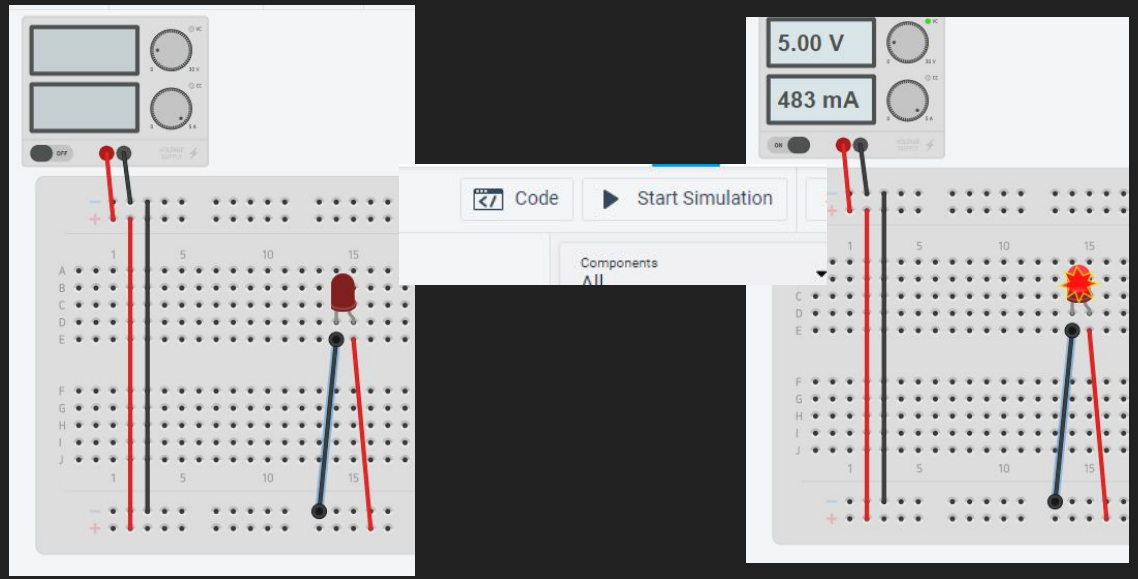


Voltage, Power Supplies, Short Circuits and LEDS

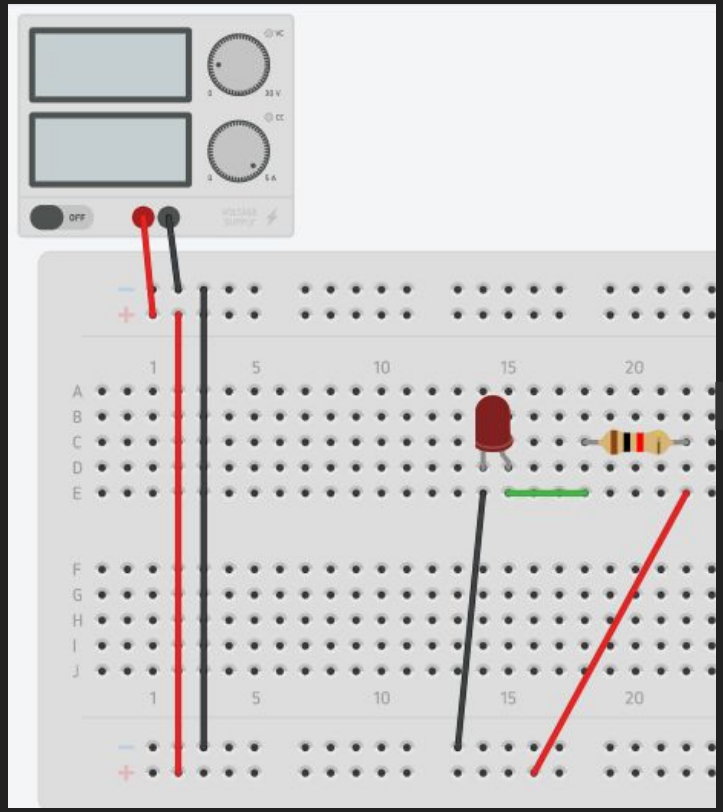


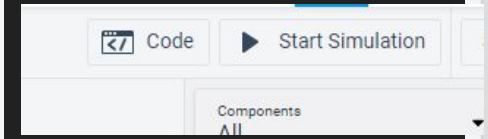


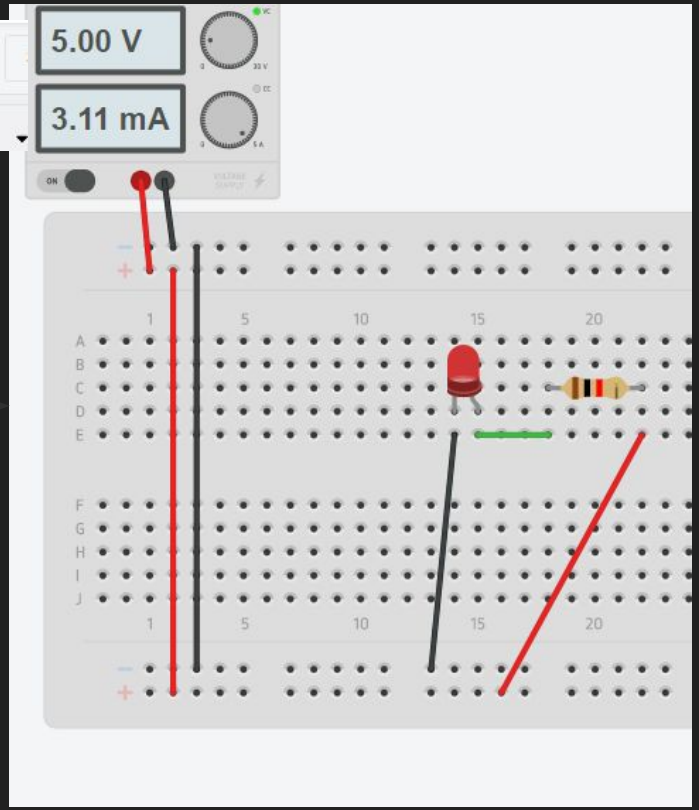




Avoiding a Short Circuit







Questions and Exercises

1. List and describe the two types of current flow.

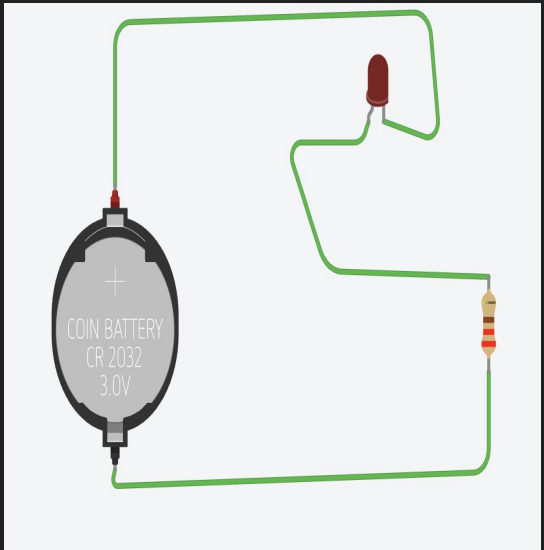
1.Conventional Current Flow Assumes current flows from the **positive (+)** terminal to the **negative (–)** terminal.

2.**Electron Flow** Describes the actual physical movement of electrons.

2. Which current flow theory is most commonly used? Conventional current flow
3. What does anode and cathode refer to?**Anode**: The terminal where conventional current *enters* a device.**Cathode**: The terminal where conventional current *exits* a device.
4. How can you tell which side of an LED is the anode and which is the cathode side?**Long lead = Anode (+) Short lead = Cathode (–)**
5. Relate voltage to the flow of water. Voltage is like **water pressure** in a pipe. **Higher voltage** = higher “pressure” pushing electrons. **Current** is like the actual amount of water flowing. **Resistance** is like the size of the pipe restricting flow.
6. What is a short circuit and how can we prevent it?A short circuit occurs when there is a **direct, low-resistance path** between the positive and negative sides of a power source. This causes **excessive current** to flow.Use **insulated wires**. Ensure components are **properly spaced**.
7. What are some common dangers of creating a short circuit? (Go to 3:35 of this video (https://youtu.be/Yo6JI_bzUzo?t=220) to see what happens when current is not limited by a resistor)**Overheating of wires ,Melting insulation ,Burns or fire hazards**
8. What unit is used to measure resistance?ohms (Ω)
9. What should be the typical resistance of a resistor connected to an LED?**220 Ω to 470 Ω** for 5V systems (very typical for Arduino circuits)

1 kΩ for lower-brightness LED indication circuits **330 Ω** is probably the most commonly used general value

10. Create a circuit with an LED and make sure the LED turns on. Reverse the LED and turn on the circuit. What happens and why? Embed a screenshot of your circuit below. Insert a link to your circuit here as well.



When the LED is connected correctly, it turns on. When I reverse the LED, it does not light up because an LED only allows current to flow in one direction.

https://www.tinkercad.com/things/eyiJ9lyGSo3-terrific-krunk-kup

11. Experiment with your last circuit and dial up the voltage on your power supply until the LED explodes/burns out. What was the voltage that caused problem? 6.5v

12. Try leaving your power supply at 5V. Now keep lowering the resistance in your resistor. At what point does your LED explode/burn out?

At **5V**, the LED burns out when the resistor is **below about 50 Ω**, because the current becomes too high.

13. Why does increasing the voltage or lowering the resistance cause the same problems as a short circuit as discussed in this lesson?

Because increasing the voltage or lowering the resistance both make the **current become too high**.

When too much current flows, the LED cannot handle it and it fails, just like in a short circuit.

14. Embed a schematic diagram of your circuit. Identify the resistor, LED, power source and ground.

